

Matthijs van der Lende

Groningen, Netherlands

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My research interests are broadly on how machine-learning systems learn, adapt, and make decisions. My specific interests include continual learning, reinforcement learning, probabilistic methods, and mechanistic interpretability.

Selected Highlights

2026	Winner, AAMAS 2026 NS-Gym Competition , with a PPO-based agent ranking first across all four evaluation criteria for adaptation to non-stationary environments.
2026	2nd Place and Scalability Award, Aletheia's Quest , LLM deception-detection competition with SAIN Groningen.
2026	Recognized as an Outstanding Reviewer, NLDL 2026.
2025	Spotlight, NeurIPS 2025 Mechanistic Interpretability Workshop for work on interpretable feature steering for preference optimization.
2025	First-author publication, NLDL 2025 , on Gaussian-process function approximation for reinforcement learning.

Selected Publications

[1] Jeremias Ferrao, **Matthijs van der Lende**, Ilija Lichkovski, and Clement Neo. “The Anatomy of Alignment: Decomposing Preference Optimization by Steering Sparse Features.” In: *NeurIPS 2025 Mechanistic Interpretability Workshop, Spotlight*. 2025.

[2] **Matthijs van der Lende**, Matthia Sabatelli, and Juan Cardenas-Cartagena. “Interpretable Function Approximation with Gaussian Processes in Value-Based Model-Free Reinforcement Learning.” In: *Proceedings of the 6th Northern Lights Deep Learning Conference (NLDL)*, Proceedings of Machine Learning Research, vol. 265, pp. 141–154. 2025.

Education

Sep 2024 – Present	M.Sc. Artificial Intelligence, University of Groningen. Master’s thesis on optimization-based continual learning, supervised by Guido van de Ven.
Sep 2020 – 2024	B.Sc. Artificial Intelligence, University of Groningen. Bachelor’s thesis: <i>Interpretable Function Approximation with Gaussian Processes in Value-Based Model-Free Reinforcement Learning</i> . The resulting work was published at NLDL 2025.

Research Experience

Jul 2025 – Mar 2026	Research Fellow, Apart Research <i>Remote</i> Conducted research in mechanistic interpretability and co-authored work presented as a spotlight at the NeurIPS Mechanistic Interpretability Workshop.
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Industry Experience

Jul – Sep 2025	Data Science Intern, Novigenix SA <i>Lausanne, Switzerland</i> Developed a text-to-SQL system for natural-language querying of internal biomedical databases and deployed the system on AWS.
Jul – Sep 2024	Machine Learning Intern, Nuwa Innovation <i>Groningen, Netherlands</i> Worked on handwriting reconstruction from smart-pen IMU data. Developed 2D displacement estimation methods using Extended Kalman Filtering and deep learning.

Teaching Experience

Sep 2021 – Present	Teaching Assistant, University of Groningen <i>Groningen, Netherlands</i> Taught 150+ students on Machine Learning, Deep Learning, Reinforcement Learning, AI Engineering, Trustworthy and Responsible AI. Gave lectures, led tutorials and lab sessions and supervised research and industry-collaborative student projects. Nominated for Teaching Assistant of the Year (2021/2022).
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Academic Service

Oct 2024 – Oct 2026	Programme Committee, University of Groningen. Master’s student member for BSc/MSc Artificial Intelligence and MSc Computational Cognitive Science; Curriculum Committee member (Oct 2024–Oct 2026) and student chair (Sep 2025–Oct 2026).
2025 – 2026	Reviewer for NLDL 2026 and the EIML Workshops at ICML 2026 and NeurIPS 2025.